

Kai Po Wei

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EDUCATION

B.S. in Vehicle Engineering

Department of Vehicle Engineering, National Taipei University of Technology, 09/2021-01/2026

- Last 60 credits: 3.9/4.0
- GPA: 3.62/4.0
- IELTS: 7.5, GRE: 321 (Q170)

RESEARCH EXPERIENCE

Structural & Optimization Lab, National Taipei University of Technology

Advisor: Prof. Hsiu-Ying Hwang

Team Lead, 09/2023 - 07/2024

Topic: *Scooter Frame Fatigue Life Analysis and Optimization*

- **Second Place**, Departmental Capstone Project Competition
- **Led a four-member team** conducted fatigue analysis on a scooter frame under five loading conditions in accordance with *industry standards*
- Developed CAD models in SolidWorks and performed finite element simulations using HyperWorks
- Evaluated stress distribution and fatigue life, optimizing the frame design for durability (fatigue life increased by 339% with only 12.5% weight increase)
- Presented findings in THE 29th NATIONAL CONFERENCE ON VEHICLE ENGINEERING

PROJECT EXPERIENCE

Taipei Tech Racing FSAE team, National Taipei University of Technology

TTR7 Race Car

Chassis team member, 10/2022 - 07/2023

- **Second Place**, Formula Student Taiwan 2023
- Designed and optimized suspension geometry to enhance vehicle *handling & stability*
- Designed, analyzed, and manufactured drivetrain components; achieved **31% cost reduction** and **16% weight savings**
- Performed finite element analysis (FEA) to ensure structural integrity of drivetrain parts
- Supported CNC machining and welding-based manufacturing

AI CUP 2025 – Ministry of Education National Intercollegiate Artificial Intelligence Competition (Taiwan)

Advisor: Prof. Tung-Ju Hsieh

Independent Researcher, 9/2025 - 1/2026

- Built end-to-end deep learning pipelines for cardiac CT analysis, achieving **0.7439 Dice** in myocardial segmentation (UNet3D) and **0.9396 Dice** in aortic valve detection (YOLOv12)
- Optimized model performance through 3D preprocessing, ROI extraction, hyperparameter tuning, and resume training

WORK EXPERIENCE

IPE Innotech Performance Exhaust

Project & Product management, R&D Intern, 07/2024 - 08/2024

- Contributed to the design, simulation testing, and manufacturing of three magnesium alloy wheels for Porsche GT3RS
- Led **five product-launch projects** and handed off three completed products to colleagues
- Increased forged wheel *production efficiency* by **16%** through *workflow redesign*

Foxconn CCEV (Center of Commercial Electric Vehicle)

R&D Intern, 07/2023-08/2023

- Participated in the design and manufacturing of a full-electric commercial van- *Model N*
- Contributed to cockpit human-factors engineering; **two proposals adopted into final design**
- Collaborated with cross-functional teams (*ADAS/Chassis/Human Factors*)

Xin-Tai cars repair shop & Horng-Yi motorcycles repair shop

Vehicles & Scooters Repair and Maintenance Apprentice, *07/2022 - 08/2022 & 07/2021 - 08/2021*

- Performed maintenance on motorcycles and vehicles
- Used OBD (On-Board Diagnostics) to diagnose problems on around five vehicles and 8 motorcycles
- Assisted in two major engine overhauls, gaining hands-on experience in engine disassembly, inspection, and reassembly (BMW 740i & Mercedes-Benz C250)

SKILLS

- **Software:** SolidWorks, HyperWorks, BikeSim, CarSim
- **Programming:** Python, MATLAB, Simulink (*co-simulation*)
- **Engineering:** *Fatigue analysis*, FEA, structural optimization, vehicle dynamics
- **Hands-on:** Mechanical maintenance, repair, machining
- **Project:** Basic project management